

A PROJECT ON

**Air Quality Index Forecasting Using
Machine Learning Techniques**

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Abstract

Air pollution has become a critical environmental and public health concern, making accurate forecasting of the Air Quality Index (AQI) essential for effective air quality management and early warning systems. This project focuses on forecasting AQI using machine learning techniques by leveraging historical air quality and meteorological data. The study utilizes pollutant concentration parameters such as PM_{2.5}, PM₁₀, NO₂, SO₂, CO, and O₃, along with relevant weather variables, to model and predict future AQI values.

Three machine learning models—Random Forest Regressor, XGBoost Regressor, and Long Short-Term Memory (LSTM) neural networks—are implemented and evaluated to capture both non-linear relationships and temporal dependencies in the data. Feature engineering techniques, including data normalization and time-lagged variables, are applied to enhance predictive performance. The models are trained and tested using historical AQI datasets obtained from publicly available sources, and their performance is compared using standard evaluation metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R² score.

The experimental results demonstrate that tree-based ensemble models provide robust performance for short-term AQI prediction, while the LSTM model effectively captures long-term temporal patterns in air quality trends. This comparative analysis highlights the strengths and limitations of different machine learning approaches for AQI forecasting. The proposed framework can assist policymakers and environmental agencies in proactive decision-making and contributes to the development of intelligent air quality monitoring systems.